INCH-POUND

MIL-M-44458
30 September 1992

MILITARY SPECIFICATION

MEAL MODULE, 10-PATIENT, PACKAGING OF ORAL LIQUID FEEDING POWDERS INTO CANS AND ASSEMBLY OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the unit packaging of oral liquid feeding powder into metal cans and the assembly of the Meal Module, 10-Patient. Each meal module consists of a single shipping container containing one day's complete menu (breakfast, lunch, and dinner) consisting of canned oral liquid powders, dairyshakes, fruit juice, and accessory packs to feed 10 people for three meals each.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8970

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SPECIFICATIONS

FEDERAL

A-A-1052	- Tube, Drinking (Plastic)
A-A-20001	- Spices, Ground and Whole, and Spice Blends
A-A-20041	- Salt, Table, Iodized
A-A-20135	- Sugar, Refined and Sugar, Brown
TT-C-495	- Coatings, Exterior, for Tinned Food Cans
UU-N-1650	- Napkin, Table, Paper
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-C-29	- Canned Subsistence Items, Packaging of
PPP-C-795	- Cushioning Material, Packaging (Flexible Cellular, Plastic Film) for Packaging Applications

MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-J-35049	- Juice, Orange, Instant
MIL-J-35050	- Juice, Grapefruit, Instant
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-J-43409	- Juice, Grape, Instant, Sweetened
MIL-S-44170	- Spaghetti and Meat with Sauce, Powder (For Oral Liquid Feeding)
MIL-C-44174	- Chili Con Carne, Powder (For Oral Liquid Feeding)
MIL-C-44175	- Corn, Buttered, Powder (For Oral Liquid Feeding)
MIL-H-44176	- Hominy Grits with Cheese, Powder (For Oral Liquid Feeding)
MIL-P-44182	- Potatoes, Mashed, Powder (For Oral Liquid Feeding)
MIL-D-44414	- Dairyshake Powder
MIL-C-44437	- Cheesecake, Powder (For Oral Liquid Feeding)
MIL-C-44438	- Corn and Potatoes, Creamed, Powder (For Oral Liquid Feeding)
MIL-G-44439	- Gingerbread, Powder (For Oral Liquid Feeding)
MIL-P-44440	- Potatoes, Delmonico, Powder (For Oral Liquid Feeding)
MIL-S-44441	- Squash, Buttered, Powder (For Oral Liquid Feeding)
MIL-C-44442	- Cake, Blueberry Upside-Down, Powder (For Oral Liquid Feeding)
MIL-B-44443	- Beans, Baked, Powder (For Oral Liquid Feeding)
MIL-S-44444	- Strawberry Shortcake, Powder (For Oral Liquid Feeding)
MIL-A-44445	- Apple Brown Betty, Powder (For Oral Liquid Feeding)
MIL-B-44446	- Banana-Strawberry Breakfast Drink, Powder (For Oral Liquid Feeding)
MIL-C-44447	- Cauliflower and Cheese, Powder (For Oral

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Liquid Feeding)
MIL-E-44448 - Escalloped Potatoes and Ham, Powder (For Oral Liquid Feeding)
MIL-H-44449 - Ham and Cheese Souffle, Powder (For Oral Liquid Feeding)
MIL-O-44450 - Oatmeal with Maple and Brown Sugar, Powder (For Oral Liquid Feeding)
MIL-P-44451 - Pea Casserole, Powder (For Oral Liquid Feeding)
MIL-T-44452 - Taco Grande, Powder (For Oral Liquid Feeding)
MIL-O-44453 - Omelet, Green Pepper and Ham, Powder (For Oral Liquid Feeding)
MIL-C-44454 - Chicken Barbecue, Powder (For Oral Liquid Feeding)
MIL-H-44455 - Ham and Cheese Fondue, Powder (For Oral Liquid Feeding)
MIL-P-44456 - Pork Oriental and Vegetables with Noodles, Powder (For Oral Liquid Feeding)
MIL-R-44457 - Rice, Fried, Powder (For Oral Liquid Feeding)
MIL-F-44459 - Farina with Peaches and Cream, Powder (For Oral Liquid Feeding)
MIL-G-44460 - Green Bean Casserole, Powder (For Oral Liquid Feeding)
MIL-P-44461 - Potatoes, Lyonnaise, Powder (For Oral Liquid Feeding)
MIL-C-44462 - Cake, Chocolate Mocha, Powder (For Oral Liquid Feeding)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF AGRICULTURE (USDA)

United Stated Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

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Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATIONS, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1120, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Components and materials.

3.1.1 Food product. The food product to be packaged shall be as specified in the applicable document (see table I).

3.1.2 Can requirements. The food product shall be unit packed into cans in accordance with level A or C as specified (see 6.2). Each can shall be filled with product in accordance with the net weight requirement of the applicable product document.

3.1.2.1 Level A. The can shall be an open-top style, round metal can, size 401 x 602, with welded side seams and compound-lined, double-seamed ends. The can shall be made throughout from not less than 0.25 pound per base box electrolytic tin plate. The base plate of the can shall be of sufficient temper to protect the product during shipment and storage. The

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can shall be coated on the outside with a coating conforming to type I or when specified type III of TT-C-495 (see 6.2). The filled can shall be hermetically sealed under a vacuum of not less than 15 inches of mercury when tested in accordance with 4.2.4. Alternatively, the can may be sealed under an atmosphere of nitrogen so that the oxygen content of the gases in the filled and sealed can shall not exceed 2.0 percent when tested in accordance with 4.2.5. When cans are closed under vacuum, the oxygen requirement is not applicable.

3.1.2.2 Level C. The cans shall be as specified in 3.1.2.1, except that cans with or without exterior commercial coating will be acceptable. Alternatively, cans may be made from 0.20 pound base box electrolytic tin plate provided the cans have an exterior commercial coating.

3.1.2.3 Plastic lid. The filled and sealed cans shall have a close-fitting plastic lid affixed securely to one end of the can. The plastic lid shall be made of reusable, self-sealing polyethylene.

3.1.2.4 Can label requirements. Cans shall be labeled in accordance with MIL-L-1497, except that only the body of the can shall be labeled. Labels shall include the following information:

PRODUCT NAME

DAY NO._____(1, 2, or 3, as applicable)

MENY TYPE_____(Breakfast, Lunch, or Dinner, as applicable)

DRY PRODUCT MUST BE USED WITHIN 30 DAYS AFTER OPENING OR DISCARD

DATE OPENENED:_____ .

INGREDIENTS (to be listed in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder)

DIRECTIONS FOR USE (as specified below)

NET WEIGHT

SERVINGS PER CONTAINER 10 (For Cake, Chocolate Mocha; Gingerbread; and Cheesecake; the label shall state: SERVINGS PER CONTAINTER: 5)

CONTRACTOR'S NAME AND ADDRESS

DATE OF PACK 1/

GAS PACKAGED (if applicable)

1/ Each can shall have the date of pack noted by using a four digit code beginning with the final digit of the current year followed by the three digit Julian day code. (For example, March 19, 1991 would read 1079.) The Julian day code shall represent the date the product is unit packed into the can.

a. For Banana-Strawberry Breakfast Drink; and Strawberry Shortcake; the following directions for use shall be specified:

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DIRECTIONS FOR USE:

1. Measure 1/2 cup powder (dry measure).
2. Place contents into a food blender.
3. Add 6 ounces cold water or milk.
   (For a pureed diet, use 3 ounces of liquid.)
4. Blend for 30 seconds on high speed.
5. Consume products within 2 hours of preparation or refrigerate.

b. For Apple Brown Betty; and Cake, Blueberry Upside Down; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Measure 2/3 cup powder (dry measure).
2. Place contents into a food blender.
3. Add 6 ounces cold water or milk.
   (For a pureed diet, use 3 ounces of liquid.)
4. Blend for 30 seconds on high speed.
5. Consume products within 2 hours of preparation or refrigerate.

c. For Cake, Chocolate Mocha; Gingerbread; and Cheesecake; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Measure 1 cup powder (dry measure).
2. Place contents into a food blender.
3. Add 6 ounces cold water or milk.
   (For a pureed diet, use 3 ounces of liquid.)
4. Blend for 30 seconds on high speed.
5. Consume products within 2 hours of preparation or refrigerate.

d. For Chicken Barbecue; Chili Con Carne; Ham and Cheese Fondue; Omelet, Green Pepper and Ham; Ham and Cheese Souffle; Spaghetti and Meat with Sauce; and Taco Grande; the following directions for use shall be specified:

1. Measure 1/2 cup powder (dry measure).
2. Place contents into a food blender.
3. Add 6 ounces hot water or milk (not less than 140^0F).
   (For a pureed diet, use 3 ounces of liquid.)
4. Blend for 30 seconds on high speed.
5. Consume products within 2 hours of preparation or refrigerate.

e. For Hominy Grits with Cheese; and Pork Oriental and Vegetables with Noodles; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Measure 1/3 cup powder (dry measure).
2. Place contents into a food blender.
3. Add 6 ounces hot water or milk (not less than 140^0F).
   (For a pureed diet, use 3 ounces of liquid.)
4. Blend for 30 seconds on high speed.
5. Consume products within 2 hours of preparation or refrigerate.

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f. For Cauliflower and Cheese; Potatoes, Delmonico; Potatoes, Lyonnaise, and Squash, Buttered; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Measure 1/4 cup powder (dry measure).
2. Place contents into a food blender.
3. Add 8 ounces hot water or milk (not less than 140^0F)
   (For a pureed diet, use 4 ounces of liquid)
4. Blend for 30 seconds on high speed.
5. Consume products within 2 hours of preparation or refrigerate.

g. For Green Bean Casserole; Corn and Potatoes, Creamed; Escalloped Potatoes and Ham; Farina with Peaches and Cream; and Potatoes, Mashed; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Measure 1/3 cup powder (dry measure).
2. Place contents into a food blender.
3. Add 8 ounces hot water or milk (not less than 140^0F)
   (For a pureed diet, use 4 ounces of liquid.)
4. Blend for 30 seconds on high speed.
5. Consume products within 2 hours of preparation or refrigerate.

h. For Beans, Baked; Pea Casserole; and Rice, Fried; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Measure 1/2 cup powder (dry measure).
2. Place contents into a food blender.
3. Add 8 ounces hot water or milk (not less than 140^0F).
   (For a pureed diet, use 4 ounces of liquid.)
4. Blend for 30 seconds on high speed.
5. Consume products within 2 hours of preparation or refrigerate.

i. For Corn, Buttered; and Oatmeal with Maple and Brown Sugar; the following directions for use shall be specified:

DIRECTIONS FOR USE:

1. Measure 2/3 cup powder (dry measure).
2. Place contents into a food blender.
3. Add 8 ounces hot water or milk (not less than 140^0F).
   (For a pureed diet, use 4 ounces of liquid.)
4. Blend for 30 seconds on high speed.
5. Consume products within 2 hours of preparation or refrigerate.

3.1.3 Menu components. The components required for menu assembly shall be as specified in table I.

TABLE I. Menu components

Components	Document reference
Oral liquid feeding powders:	

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Apple brown betty	MIL-A-44445
Beans, baked	MIL-B-44443
Banana-strawberry breakfast drink	MIL-B-44446
Cake, blueberry upside-down	MIL-C-44442
Cake, chocolate mocha	MIL-C-44462
Cauliflower and cheese	MIL-C-44447
Cheesecake	MIL-C-44437
Chicken barbecue	MIL-C-44454
Chili con carne	MIL-C-44174
Corn, buttered	MIL-C-44175
Corn and potatoes, creamed	MIL-C-44438
Escalloped potatoes and ham	MIL-E-44448
Farina with peaches and cream	MIL-F-44459
Gingerbread	MIL-G-44439
Green bean casserole	MIL-G-44460
Ham and cheese fondue	MIL-H-44455
Ham and cheese souffle	MIL-H-44449
Hominy grits with cheese	MIL-H-44176
Oatmeal with maple and brown sugar	MIL-O-44450
Omelet, green pepper and ham	MIL-O-44453
Pea casserole	MIL-P-44451
Pork oriental and vegetables with noodles	MIL-P-44456
Potatoes, delmonico	MIL-P-44440
Potatoes, lyonnaise	MIL-P-44461
Potatoes, mashed	MIL-P-44182
Rice, fried	MIL-R-44457
Spaghetti and meat with sauce	MIL-S-44170
Squash, buttered	MIL-S-44441
Strawberry shortcake	MIL-S-44444
Taco grande	MIL-T-44452
Dairyshakes:	
Dairyshake, banana	MIL-D-44414
Dairyshake, chocolate	MIL-D-44414
Dairyshake, eggnog	MIL-D-44414
Dairyshake, orange	MIL-D-44414
Dairyshake, strawberry	MIL-D-44414
Dairyshake, vanilla	MIL-D-44414
Fruit juices:	
Juice, grape, instant	MIL-J-43904
Juice, grapefruit, instant	MIL-J-35050
Juice, orange, instant	MIL-J-35049

3.1.4 Accessory pack. The components indicated in table II shall be inserted into a 3 1/2 inch wide by 9 1/2 inch long, flat style, plastic bag made from minimum thickness 0.002 inch clear, food grade polyethylene film. The filled bag shall be closed by either heat or electronic sealing. Prior to the sealing operation, excess air shall be expelled from the bag. Ten bagged accessory packs shall then be banded with a rubber band.

TABLE II. Accessory pack contents

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Item	Quantity	Document reference
Drinking tube	1	A-A-1052, type II
Napkin	1	UU-N-1650, type II, class 2
Pepper	1	A-A-20001, .15 gram envelope
Salt	1	A-A-20041, .75 gram packet
Sugar	1	A-A-20135, type I, class 1 or 2, 6.0 gram heat sealed envelope

3.2 Menu assembly. The components listed in table III shall be packed as specified in 5.1.

TABLE III. Menu contents

Meal	Day No., components therein	Unit issue (401 x 602 can)
	Day 1	
Breakfast	Banana-strawberry breakfast drink	1
	Oatmeal with maple and brown sugar	1
	Juice, orange	1
	Dairyshake, vanilla	2
Lunch	Ham and cheese souffle	1
	Pea casserole	1
	Potatoes, mashed	1
	Apple brown betty	2
	Dairyshake, strawberry	2
Dinner	Chili con carne	1
	Cauliflower and cheese	1
	Escalloped potatoes and ham	1
	Cake, blueberry upside-down	2
	Dairyshake, chocolate	2
	Day 2	
Breakfast	Hominy grits with cheese	1
	Omelet, green pepper and ham	1
	Juice, grapefruit	1
	Dairyshake, orange	2
Lunch	Taco grande	1
	Squash, buttered	1
	Potatoes, delmonico	1
	Strawberry shortcake	1
	Dairyshake, eggnog	2
Dinner	Chicken barbecue	1
	Corn, buttered	2
	Beans, baked	1
	Cake, chocolate mocha	2
	Dairyshake, banana	2

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Day 3

Breakfast	Farina with peaches and cream	1
	Ham and cheese fondue	1
	Juice, grape	1
	Dairyshake, vanilla	2
Lunch	Spaghetti and meat with sauce	1
	Green bean casserole	1
	Potatoes, lyonnaise	1
	Gingerbread	2
	Dairyshake, strawberry	2
Dinner	Pork oriental and vegetables with noodles	1
	Corn and potatoes, creamed	1
	Rice, fried	1
	Cheesecake	2
	Dairyshake, chocolate	2

3.3 Plant Qualification. The product shall be packaged and assembled in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food," and the plant sanitation requirements of the appropriate Government inspection agency.

3.4 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA or other government inspection agency shall not relieve the contractor of the obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA or other government inspection agency for any inspection.

4.1.1 Inspection and certification. Product acceptability shall be determined by the USDA or other government inspection agency. The USDA or other government inspection agency will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1 Unfilled can inspection. Conformance of unfilled cans to the requirements in 3.1.2 shall be determined by examination of certificates of

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conformance or of other valid documents. Any nonconformance shall be cause for rejection of the can lot or of any involved product.

4.2.2 In-process examination. In-process examination shall be performed to determine conformance to the filling, sealing and packaging requirements. Any nonconformance revealed by actual examination or by review of records or of other valid documents shall be cause for rejection of the involved product.

4.2.3 Can condition examination. Examination of filled and sealed cans shall be in accordance with the United States Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with MIL-L-1497 (see 3.1.2.4).

4.2.4 Leakage and vacuum (when applicable) examination. Examination for leakage and vacuum shall be in accordance with PPP-C-29 (see 3.1.2.1).

4.2.5 Oxygen content testing (when applicable). Eight filled and sealed cans shall be selected at random from the lot. The filled and sealed cans shall be tested for oxygen in headspace by using an electronic oxygen analyzer which operates on the principle of the difference in partial pressure of oxygen between the oxygen reference and the oxygen content of the sample as detected by a porous ceramic Zirconia sensor, such as the Illinois Instruments analyzer or its equivalent; or on the principle of paramagnetic resonance such as the Servomex analyzer, or its equivalent. The oxygen analyzer shall be calibrated to a known standard prior to testing the headspace gas of the product. Test results shall be reported to the nearest 0.1 percent for each can. Any nonconformance to the requirement specified in 3.1.2.1 shall be classified as a major defect and shall be cause for rejection of the lot.

4.2.6 Accessory pack examination. The filled and sealed accessory packs shall be examined for the defects listed in table IV. The lot size shall be expressed in packs. The sample unit shall be one filled and sealed pack. The inspection level shall be S-4 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE IV. Accessory pack defects

Category		Defect
Major	Minor	
101		Size and style not as specified.
102		Not sealed as specified.
103		Missing, torn, or crushed straw.
104		Not clean (accessory pack or outer packaging of any of its contents). 1/
	201	Tear, hole, or open seam.
	202	Excess air remaining in bag after final closure.
	203	Missing (except straw) or unserviceable component (for

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example, tear, hole, or open seam in pepper, salt, or sugar).

1/ Outer packaging shall free from foreign matter which is unwholesome, has the potential to cause package damage (for example, glass, metal filings, etc.), or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential package damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Dried product which effects less than 1/8 of the total surface area of one package face (localized and aggregate).

4.2.7 Shipping container examination (for meal module). Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.

Minor: Other required markings missing, incorrect, or illegible.
Not waterproofed by means of tape.
Container not snug-fitting.

4.2.8 Meal module interior examination. The assembled meal modules shall be examined for the defects listed in table V. The lot size shall be expressed in meal modules. The sample unit shall be one filled and closed meal module. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE V. Meal module defects

Category		Defect
Major	Minor	
101		Care leaking or damaged (so as to expose contents).
102		Components missing or not as specified.
103		Components not packed or arranged as specified.
104		Interior packing, air cell, or cushioning not as specified.
	201	Tear, hole, or open seam in accessory pack.
	202	Accessory packs not banded in groups of ten.

4.2.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

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5.1 Packing. The meal module shall be packed in accordance with level A, B, or C as specified (see 6.2).

5.1.1 Level A packing. One day's complete menu (breakfast, lunch, and dinner) as specified in table III, and 30 accessory packs, as specified in 3.1.4, shall be packed into a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. The inside dimensions of the shipping container shall be 16-1/2 inches in length, 12-1/4 inches in width, and 12-3/8 inches in depth. The canned products shall be arranged in two tiers with the bottom tier consisting of 4 cans in length, and 3 cans in width. The cans shall be packed on end with the plastic lid on top. The top tier shall be fitted with an open corrugated cell measuring 12-1/4 inches in length, 8-1/4 inches in width, and 6-5/16 inches in depth, fabricated from V3c fiberboard. The open cell shall be positioned in the middle of the top tier with the long dimension spanning the width of the box. The remaining cans of product shall be placed in the spaces outside this cell except that one can of product remaining shall be placed inside the cell. Thirty accessory packs shall be placed inside the cell. Cushioning material conforming to PPP-C-795 shall be used as necessary to fill voids and immobilize the contents of each shipping container to prevent movement or damage during transit. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.1.2 Level B packing. One day's complete menu (breakfast, lunch, and dinner), as specified in table III, and 30 accessory packs, as specified in 3.1.4, shall be packed as specified in 5.1.1, except the shipping container shall be grade V3c, V3s, or V4s of PPP-B-636.

5.1.3 Level C packing. One day's complete menu (breakfast, lunch, and dinner), as specified in table III, and 30 accessory packs, as specified in 3.1.4, shall be packed as specified in 5.1.1, except the shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable, except the closure of the fiberboard shipping container shall be in accordance with method II as specified in appendix of PPP-B-636.

5.2 Unit loads. When specified (see 6.2), the product, packed as specified in 5.1.2 or 5.1.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.3 Marking.

5.3.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following information shall be marked on each shipping container:

DAY NO._____(1, 2, or 3 as applicable)

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(Includes breakfast, lunch, and dinner components and accessory packs for 10 patients.)

5.3.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The products are intended to be used by patients requiring an oral liquid or pureed diet (for example, those with maxillofacial injuries, wired jaws, oral surgery, dental diseases, or swallowing difficulties).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type of cans required (see 3.1.2).
d. When type III exterior can coating is required (see 3.1.2.1).
e. Level of packing required (see 5.1).
f. Type and class of unit load when unit loading is required (see 5.2).

6.3 Subject term (key word) listing.

Combat field feeding
Dental diseases
Hospital feeding
Jaw injury
Liquid diet
Maxillofacial injury
Operational ration
Pureed
Shelf stable

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8970-0169)

Review activities:

Army - MD, QM
Navy - MC
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-44458	1992 September 30

3. DOCUMENT TITLE

MEAL MODULE, 10-PATIENT, PACKAGING OF ORAL LIQUID FEEDING POWDERS INTO CANS AND ASSEMBLY OF

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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